

Mathematics A

Unit 2 • Chapter OL-T33 Classified

Cross-session • chapter-organised candidate

5 classified items • 17 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Unit 2 • Chapter OL-T33 • 5 items • 17 marks

Chapter	Items	Marks	Starts
01. OL-T33 Graphs of Functions	5	17	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Graphs of Functions

Unit 2 • 17 marks • 5 item(s)

November 2024 | 4MA1-1H Q23 | 5 marks

Infer a graph parameter from two intersections

January 2022 | 4MA1-2H Q21 | 2 marks

Estimate roots from axis crossings

January 2022 | 4MA1-2H Q21 | 3 marks

Draw a line to solve a nonlinear equation

January 2023 | 2H Q24 | 2 marks

Draw a line to solve a nonlinear equation

November 2025 | 2H Q12 | 5 marks

Draw a line to solve a nonlinear equation

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Original question 15

4MA1-1H Q23 M01

MODULAR UNIT 2

5 marks

- 23 The curve **C** has equation $y = x^2 - 8x - 9$
The straight line **L** has equation $y = k$ where k is an integer.

C and **L** intersect at the points A and B

The coordinates of point A are (p, k)

The coordinates of point B are (q, k)

Given that $p - q = 14$

find the value of k

Show clear algebraic working.

Original question 15 (continued)

4MA1-1H Q23 M01

MODULAR UNIT 2

5 marks

 $k =$

Worked solution 15Use the chord separation to find k

MODULAR UNIT 2

5 marks

M01 – Use the chord separation to find k **Use the equal y -values**

$$p^2 - 8p - 9 = k \text{ and } q^2 - 8q - 9 = k.$$

Use symmetry

The axis is $x=4$, so $(p+q)/2=4$; with $p-q=14$, $p=11$, $q=-3$.

Substitute

$$k = 11^2 - 8(11) - 9 = 24.$$

Final answer

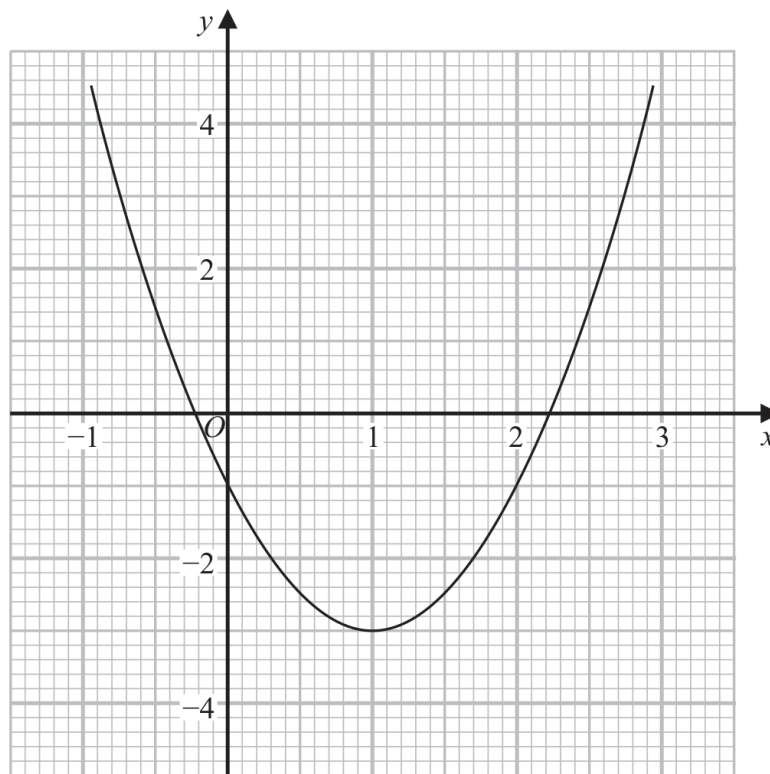
$$k = 24$$

Original question 32

4MA1-2H Q21 Q21(a)

ORIGINAL QUESTION**2 marks**

21 Part of the graph of $y = 2x^2 - 4x - 1$ is shown on the grid.



- (a) Use the graph to find estimates for the solutions of the equation $2x^2 - 4x - 1 = 0$
Give your solutions correct to one decimal place.

.....
(2)

Worked solution 32

Chapter: Graphs of Functions

MODULAR UNIT 2

2 marks

Q21(a) – Chapter: Graphs of Functions

Family: Solve using graph roots or intersections • Variant: Estimate roots from axis crossings

MethodThe x-intercepts of $y=2x^2-4x-1$ are approximately $x=-0.2$ and $x=2.2$.*Why: Read where the curve crosses the x-axis.***Method**

Both values are given to one decimal place.

*Why: Round the graph readings as requested.***Final answer****roots:** -0.2 and 2.2

Original question 33

4MA1-2H Q21 Q21(b)

ORIGINAL QUESTION**3 marks**

- (b) By drawing a suitable straight line on the grid, find estimates for the solutions of the equation $x^2 - x - 1 = 0$

Show your working clearly.

Give your solutions correct to one decimal place.

.....
(3)

Worked solution 33

Chapter: Graphs of Functions

MODULAR UNIT 2

3 marks

Q21(b) – Chapter: Graphs of Functions

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

MethodFor $y=x^2-x-1$, use $y=-2x+1$ as the straight line.*Why: Rearrange $x^2-x-1=0$ to $x^2-x=1$ and compare with the displayed quadratic.***Method**Draw $y=-2x+1$ through $(0,1)$ and $(-1,3)$.*Why: Use two points to draw the line on the supplied grid.***Method**The intersections give $x \approx -0.6$ and $x \approx 1.6$.*Why: Read the two x -coordinates to one decimal place.***Final answer**roots: -0.6 and 1.6

Completed diagram 33

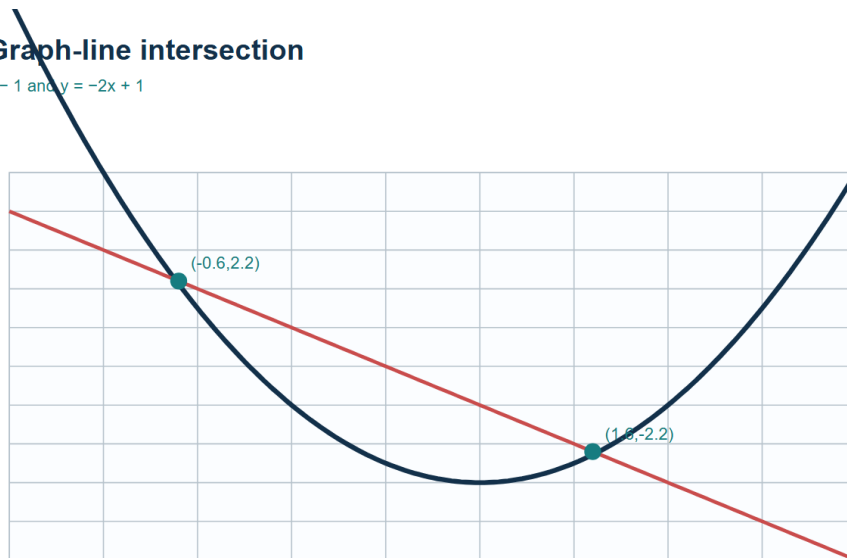
Chapter: Graphs of Functions

MODULAR UNIT 2

3 marks

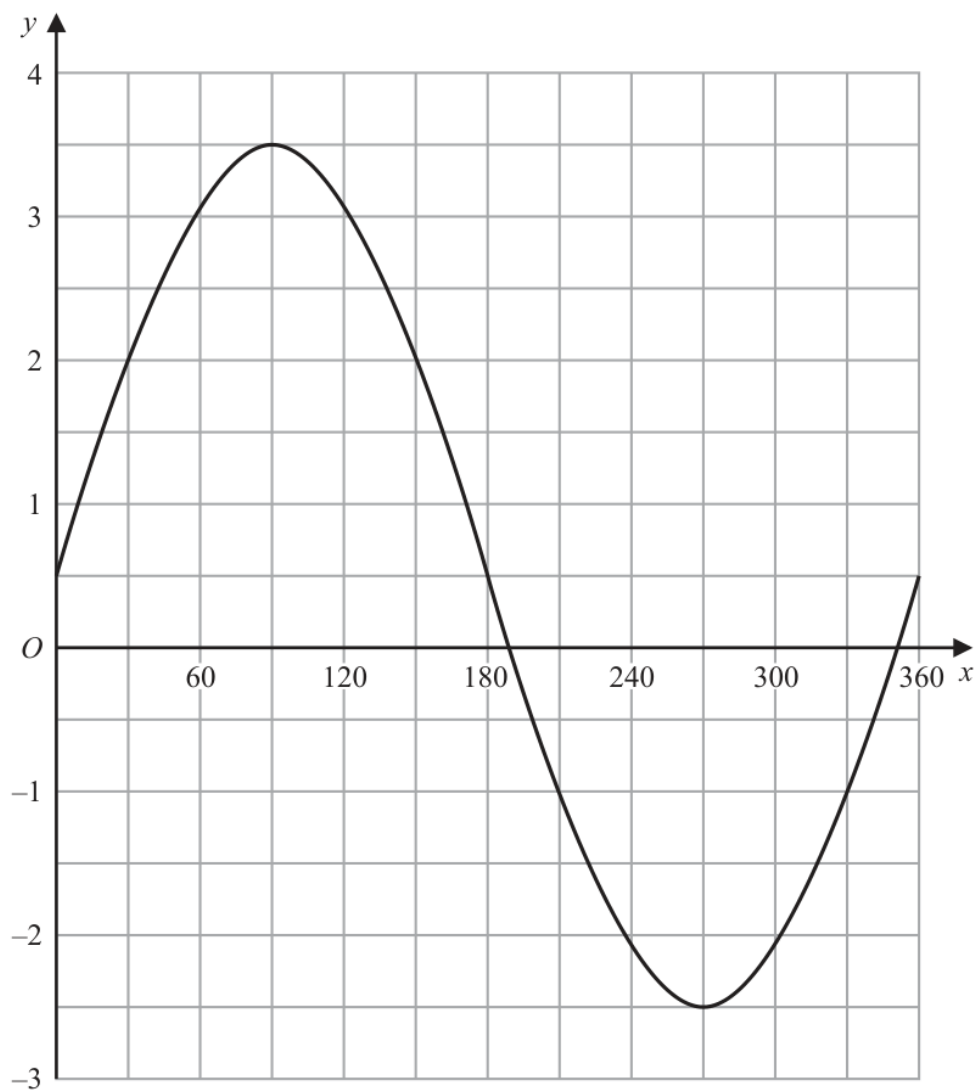
Q21 • Graph-line intersection

$$y = 2x^2 - 4x - 1 \text{ and } y = -2x + 1$$

x-intersections: -0.6 and 1.6

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24 The graph of $y = a \sin x^\circ + b$ is drawn on the grid.



Find the value of a and the value of b

$a =$

$b =$

(Total for Question 24 is 2 marks)

Answer reference | January 2023 | Question 24

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

Q	working	Answer	Mark	Notes
24		3	2	B1
		0.5		B1 oe
				Total 2 marks

Worked solution

January 2023 | Question 24 | 2 marks

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

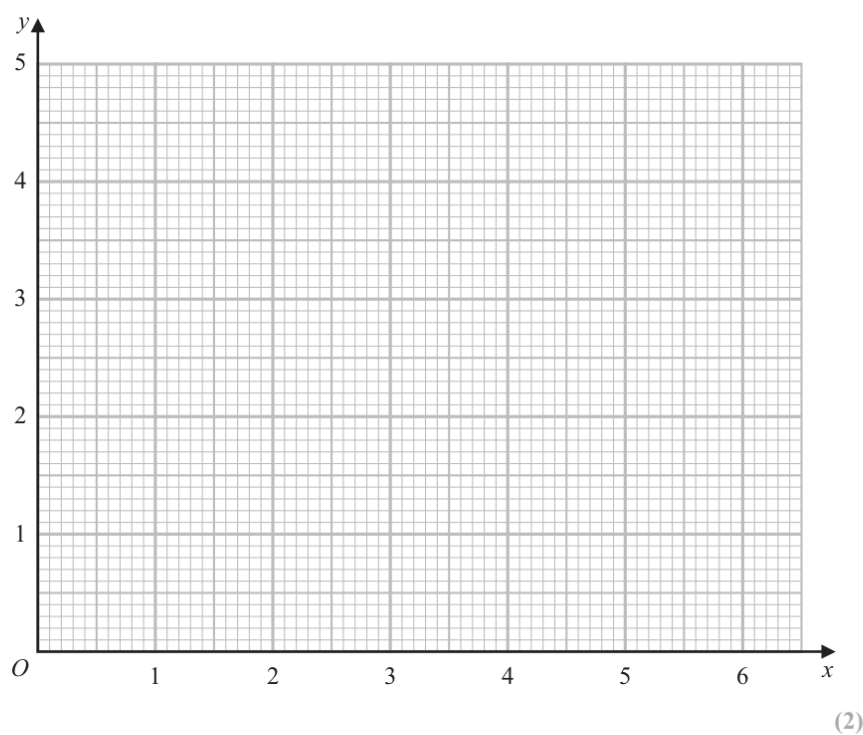
Official final mark-scheme pages are embedded immediately after this question.

12 (a) Complete the table of values for $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$

x	0.5	1	2	3	4	5	6
y	4.25			2.17	2.5	2.9	3.33

(1)

(b) Draw the graph of $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$ for $0.5 \leq x \leq 6$



(2)



- (c) By drawing a suitable line on the grid, find estimates for the solutions of the equation $x + \frac{4}{x} = 6$

Give your answers correct to one decimal place.

.....
(2)

(Total for Question 12 is 5 marks)

Worked solution

November 2025 | Question 12 | 5 marks

Family: Solve using graph roots or intersections • Variant: Draw a line to solve a nonlinear equation

Solution

$$y = \frac{1}{2}\left(x + \frac{4}{x}\right)$$

The completed table is:

$\begin{array}{c|cccccc} \end{array}$

$x \quad 0.5 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5 \quad 6 \quad \backslash$

y

$4.25 \quad 2.5 \quad 2 \quad 2.17 \quad 2.5 \quad 2.9 \quad 3.33$

$\end{array}$

For

$$x + \frac{4}{x} = 6$$

we need

$$\frac{1}{2}\left(x + \frac{4}{x}\right) = 3$$

So draw the horizontal line $y=3$ on the graph.

The intersections give approximately

$$x = 0.8 \text{ quad and quad } x = 5.2$$

Answer: Table missing values 2.5, 2. Estimates 0.8 and 5.2.